

ONTARIO, CA, USA

WMO: 747040

Lat: 34.056N

Lon: 117.600W

Elev: 289

StdP: 97.90

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 03102

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.7	5.0	-15.9	1.0	18.9	-12.5	1.3	18.0	15.0	16.9	12.5	16.9	1.2	40	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.3	38.0	20.7	36.5	20.2	34.9	19.7	22.8	34.2	21.9	33.0	21.2	31.9	5.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.3	26.6	18.0	13.4	25.4	17.5	13.0	24.5	68.3	34.2	65.3	33.2	62.4	31.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	7.8	6.9	DB	0.4	42.0	2.0	1.8	-1.0	43.3	-2.2	44.3	-3.3	45.4	-4.7	46.7	(4)
(5)			WB	-1.9	24.2	1.5	1.3	-3.0	25.1	-3.9	25.9	-4.8	26.6	-5.9	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	13.6	13.6	15.7	17.3	19.5	22.6	25.9	26.5	24.8	20.7	16.5	13.0
	DBStd	5.77	3.36	3.33	3.45	3.68	3.41	3.18	2.64	2.76	3.48	3.56	3.55	3.01
	HDD10.0	21	6	5	1	1	0	0	0	0	0	0	1	7
	HDD18.3	735	152	137	96	61	26	4	0	0	1	16	77	166
	CDD10.0	3369	117	105	178	221	294	377	494	512	445	331	195	101
	CDD18.3	1043	4	4	14	32	61	131	236	253	196	89	21	2
	CDH23.3	11228	72	93	229	419	634	1306	2472	2743	2017	919	283	42
CDH26.7	5650	9	20	68	159	270	626	1340	1550	1091	419	95	3	
Wind	WSAvg	2.7	2.3	2.3	2.7	3.0	3.2	3.1	3.1	2.9	2.6	2.5	2.1	2.2
Precipitation	PrecAvg	568	127	142	76	37	14	4	3	2	5	24	36	91
	PrecMax	1013	532	415	293	111	141	32	14	11	30	162	81	303
	PrecMin	242	9	6	0	5	0	0	0	0	0	0	2	16
	PrecStd	234	138	104	76	27	30	9	4	3	8	36	24	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	29.2	31.9	34.2	36.1	38.6	39.5	40.3	40.0	36.6	32.5	27.1
		MCWB	12.7	13.6	14.9	15.6	18.0	20.2	21.5	21.5	20.1	17.2	14.7	12.1
	2%	DB	25.4	26.3	28.6	30.9	32.5	35.1	37.0	37.8	37.4	33.8	29.6	24.2
		MCWB	11.6	12.5	14.3	15.1	17.2	19.8	21.6	21.0	20.3	16.5	14.1	11.3
	5%	DB	23.1	23.4	25.9	27.9	29.8	32.5	35.1	36.2	35.2	31.4	26.7	22.1
		MCWB	11.1	11.8	13.6	14.5	17.0	19.2	20.7	20.4	19.6	15.8	13.2	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	21.0	20.3	23.0	25.4	27.0	30.4	33.4	34.1	32.8	28.6	23.9	19.5
		MCWB	10.4	11.2	13.0	13.7	16.6	18.8	20.3	20.0	18.9	15.3	12.2	10.1
	0.4%	WB	15.5	15.9	17.7	18.5	21.0	22.4	24.2	24.4	22.9	20.8	16.9	15.2
		MCDB	20.9	23.7	26.1	29.5	31.9	33.3	36.1	35.3	35.3	28.9	23.1	19.1
	2%	WB	14.0	14.4	16.1	17.1	19.3	21.0	22.8	22.8	21.9	19.2	15.7	14.0
		MCDB	18.9	20.1	24.6	26.9	29.3	32.0	34.0	34.3	33.2	27.3	22.9	18.6
Mean Daily Temperature Range	5% DB	WB	13.0	13.4	15.1	15.7	18.1	20.1	21.8	21.8	20.9	18.1	14.9	13.0
		MCDB	17.7	19.5	22.3	24.3	27.0	30.4	32.7	33.0	31.6	26.3	22.3	17.1
	5% WB	WB	12.2	12.5	14.1	14.8	17.1	19.2	21.0	20.8	20.1	17.1	14.0	12.0
		MCDB	17.3	18.1	20.6	22.6	24.8	28.6	31.2	31.9	29.7	24.6	20.8	16.2
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.335	0.350	0.352	0.362	0.377	0.361	0.345	0.338	0.301	0.300	
	taud	2.544	2.534	2.448	2.413	2.437	2.422	2.442	2.474	2.505	2.496	2.591	2.587	
	Ebn at Noon	914	942	940	936	931	918	902	914	918	898	912	899	
	Edh at Noon	81	91	107	116	115	116	113	108	100	93	76	74	
All-Sky Solar Radiation	RadAvg	3.22	4.12	5.54	6.75	7.40	7.95	7.66	7.25	6.17	4.70	3.59	2.92	
	RadStd	0.22	0.41	0.28	0.30	0.39	0.40	0.36	0.19	0.19	0.28	0.17	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.61	N/A	-2.87	N/A	N/A	N/A	N/A	-82	+236	+138
(47)	Regional (51 neighbors)	+0.63	N/A	-1.70	+0.72	N/A	N/A	N/A	-112	+223	+112

Nomenclature: See separate page